

In the 1930s, the psychologist Edward Tolman, working at UC Berkeley, was putting rats in mazes to see how they learned. Normal psychology-type work at the time—this was the generation that followed Thorndike. The research paradigm of Thorndike’s law of effect, in which animals repeated behavior that had pleasant consequences, was in full force.

Tolman noticed something odd. When rats reached forks in his mazes where the choice was not obvious—where it wasn’t clear whether they should go left or right—rats would pause and look back and forth for a few seconds before choosing a direction. This made no sense in the standard Thorndikian view that all learning occurred through trial and error—why would the behavior of pausing and toggling one’s head back and forth have been reinforced?

Tolman made a speculation: The rat was “playing out” each option before taking it. Tolman called this “*vicarious* trial and error.”

Rats expressed this head toggling behavior only when decisions were hard. One way to make decisions hard is for the costs to be close to the benefits. Suppose you put a rat in a tunnel where it passes various doors, each of which leads to food. And suppose that when rats pass these doors, a specific sound is made that signals how long a rat will have to wait for food if it chooses to walk through the door. One sound signals that rats will have to wait only a few seconds; another sound signals that the rat will have to wait up to half a minute. Once rats learn how all this works, they do not perform their head toggling at the doors with a short delay (they just immediately run in to get the food, presumably thinking, *Obviously this is worth the wait*) or at doors with a long delay (they just immediately walk past the door: *No way this is worth the wait, I’ll just check out the next door*). But they *do* perform the head toggling behavior at the doors with a medium delay (*Is this worth the wait or should I try the next option?*).

Another way to make decisions difficult is to change the rules. If suddenly food is no longer where a rat expects it in the maze, the next time the rat is placed in the maze it will substantially increase its head toggling behavior, seeming to consider alternative paths. Similarly, suppose a rat enters a maze with two types of food, and suppose recently the rat has eaten a lot of one of those types of food (and hence no longer wants that food), then this head toggling behavior emerges (*Do I want to turn left to get X instead of turning right and getting Y?*).

Of course, the fact that you can observe a rat pause and turn its head back and forth doesn't prove that it is, in fact, imagining going down different paths. And due to this lack of evidence, the idea of vicarious trial and error faded out of favor in the decades that followed Tolman's observation. It was only recently, in the 2000s, that technology reached the point where ensembles of neurons in the brain of a rat could be recorded in real time as rats navigated their environment. This was the first time that neuroscientists could literally watch what was happening in the brains of rats when they paused and toggled their heads back and forth.

It was David Redish and his student Adam Johnson, neuroscientists at the University of Minnesota, who first probed what was happening in the brain of rats during these choices. At the time, it was well known that when a rat navigates a maze, specific place cells in its hippocampus get activated. This was similar to the spatial map of a fish—specific hippocampal neurons encode specific locations. In a fish, these neurons become active only when the fish is physically present at the encoded location—but when Redish and Johnson recorded these neurons in rats, they found something different: when the rat stopped at the decision point and turned its head back and forth, its hippocampus ceased to encode the actual location of the rat and instead went back and forth rapidly playing out the sequence of place codes that made up both possible future paths from the choice point. Redish could literally see the rat imagining future paths.

How groundbreaking this was cannot be overstated—neuroscientists were peering directly into the brain of a rat, and directly observing the rat considering alternative futures. Tolman was right: the head toggling behavior he observed was indeed rats planning their future actions.

In contrast, the first vertebrates did not plan their actions ahead of time. We can see this by examining their cold-blooded descendants—modern fish and reptiles—who show no evidence of learning by vicarious trial and error.

Consider the detour task. Take a fish and put it in a tank with a transparent barrier in the middle of the tank. Put a small hole in one corner of the barrier so that the fish can get from one side to the other. Let the fish explore the tank, find the hole, and spend some time swimming back and forth. After several days, do something new: Put the fish on one side of the tank and put a treat on the opposite side of the transparent barrier. What happens?